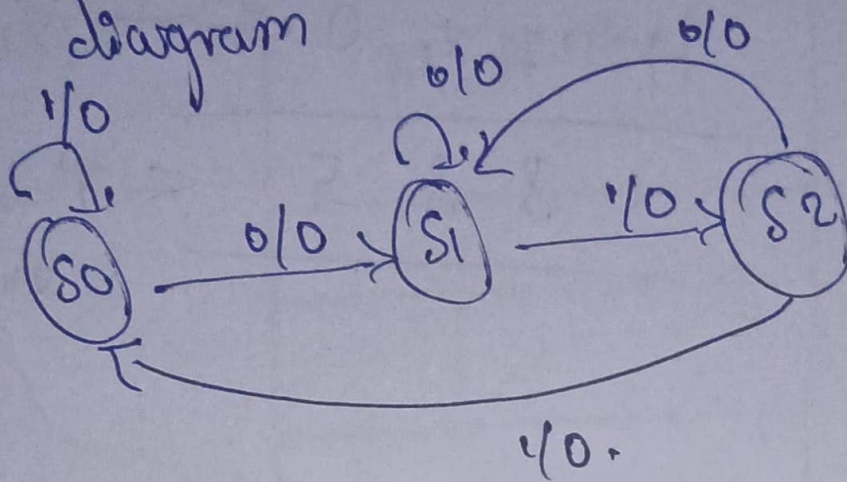


FSM Assignment

➤ Draw the state diagram for a FSM, the output toggles whenever it detects '110' LSB first (overlapping) Consider init value of output as '0'.

• state diagram



Previous state		input I	next state		output(y)
Q ₁	Q ₀		D ₁	D ₀	
S ₀ {	0	0	0	1	0
	0	1	0	0	0
	1	0	0	1	0
S ₁ {	0	1	1	0	0
	1	1	1	0	1
	0	0	0	0	0
S ₂ {	1	0	1	x	0
	1	0	x	x	0
	x	x	x	x	

Truth Table 1:

Q_1	Q_0	I	D_1	D_0
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

Truth Table 2:

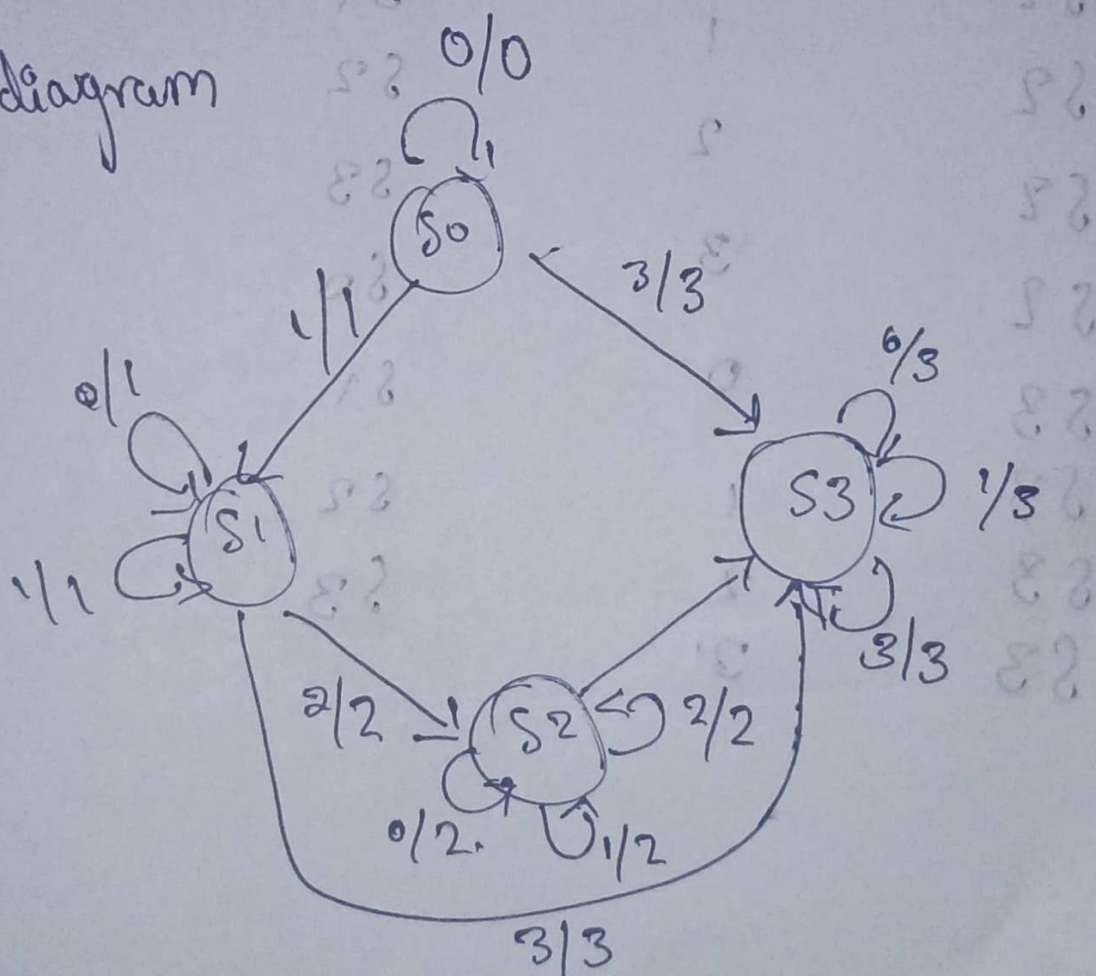
Q_1	Q_0	I	D_1	D_0
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

$\therefore D_1 = Q_0 I$

$D_0 = \bar{I}$

$D_2 = Q_1 \bar{Q}_0 I$

② Design a FSM over $\{0, 1, 2, 3\}$ that will output the largest input digit read so far.
 eg. input: 001032 will produce output: 001133.
 State diagram



Previous State Input next state

S0	0	S0
S0	1	S1
S0	2	S2
S0	3	S3
S0	0	S0
S1	1	S1
S1	2	S2
S1	3	S3
S1	0	S0
S2	1	S1
S2	2	S2
S2	3	S3
S2	0	S0
S3	1	S1
S3	2	S2
S3	3	S3
S3	0	S0

